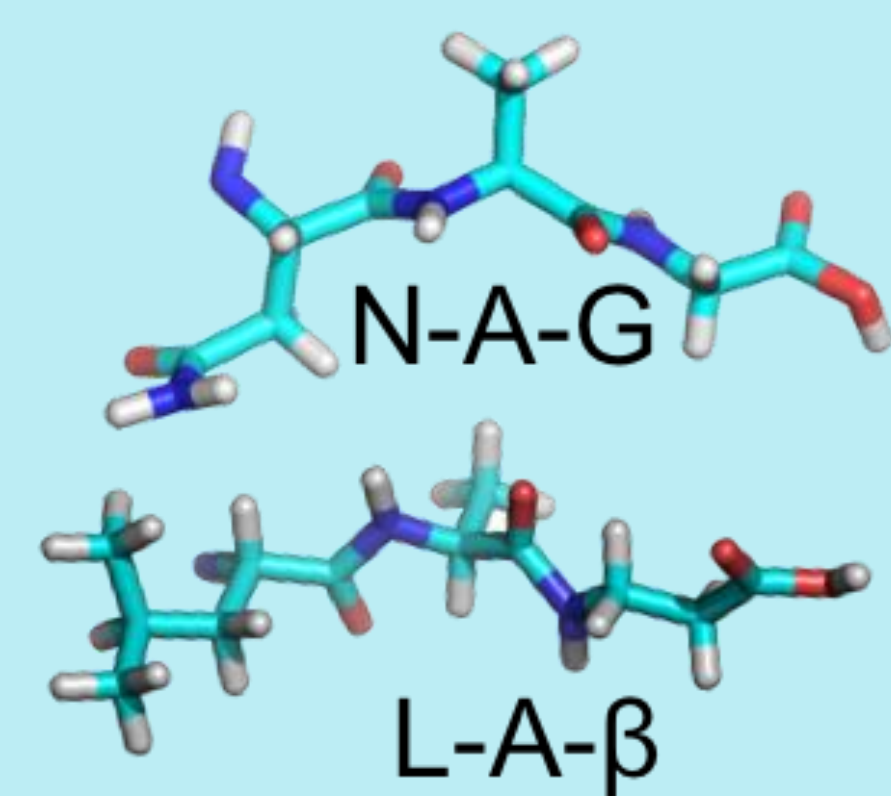




Optimization of IEDDA reaction parameters

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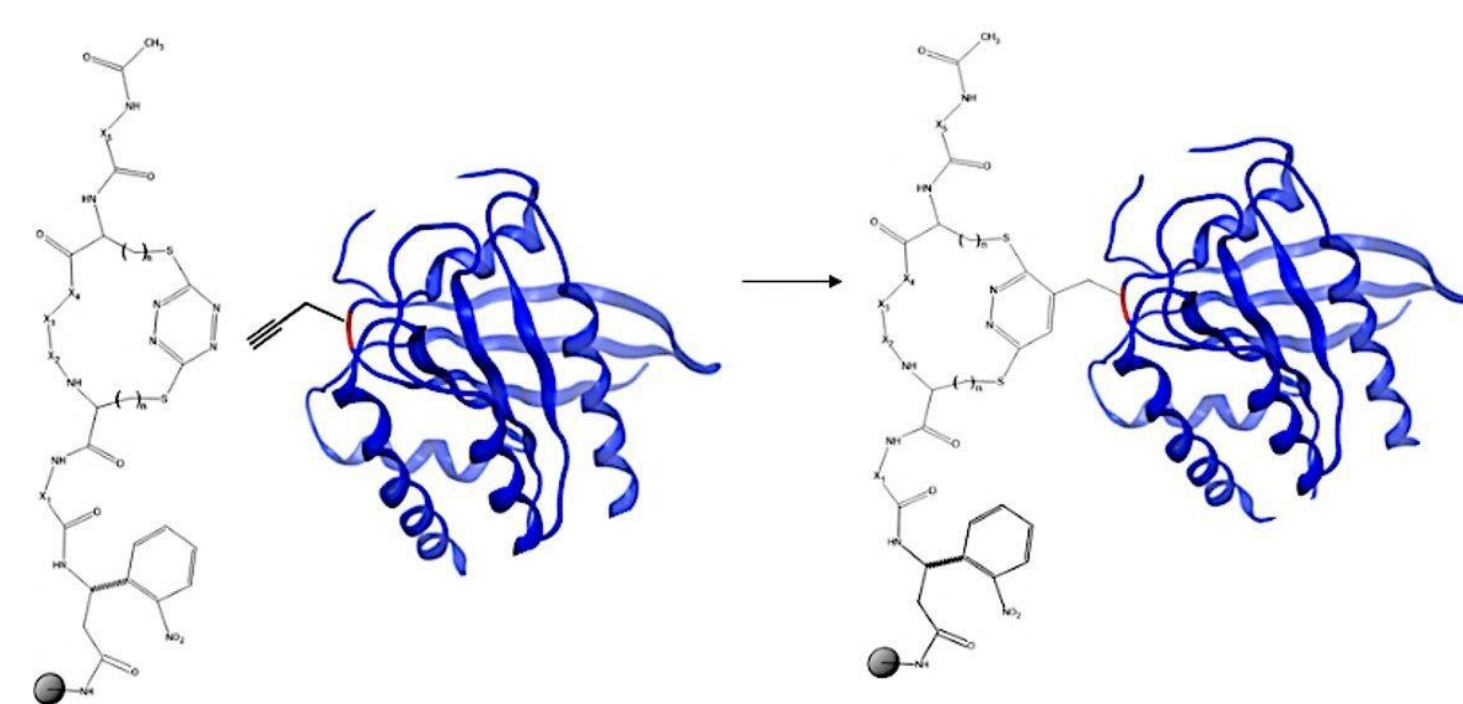
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Introduction

Inverse Electron Demands Diels-Alder (IEDDA) reaction is a biorthogonal reaction that potentially can be used for various applications such as bioconjugation, imaging, screening etc. The reaction kinetics of the IEDDA reaction varies based on the nature of the substitutions on the tetrazine and diene. Incorporation of S, S-disubstituted tetrazine in biomolecules such as peptides is synthetically much less challenging than the incorporation of an aromatic N, N-tetrazines. Developing S, S-tetrazine linked molecules in biocompatible systems can be easily synthesized. Studying their reaction kinetics with moderately reactive dienophiles like norbornene-containing or cyclopropene-containing molecules under various organic and aqueous conditions using various spectroscopic and analytical techniques have been going on.



Methods

(1) Create linear peptide using Alstra Machine

(2) Cyclization with tetrazine (Swell resin; 3% TFA + DCM; Reduction with Beta-Mercaptoethanol + DMF; Tetrazine and 2,6 lutidine; Argon gas)

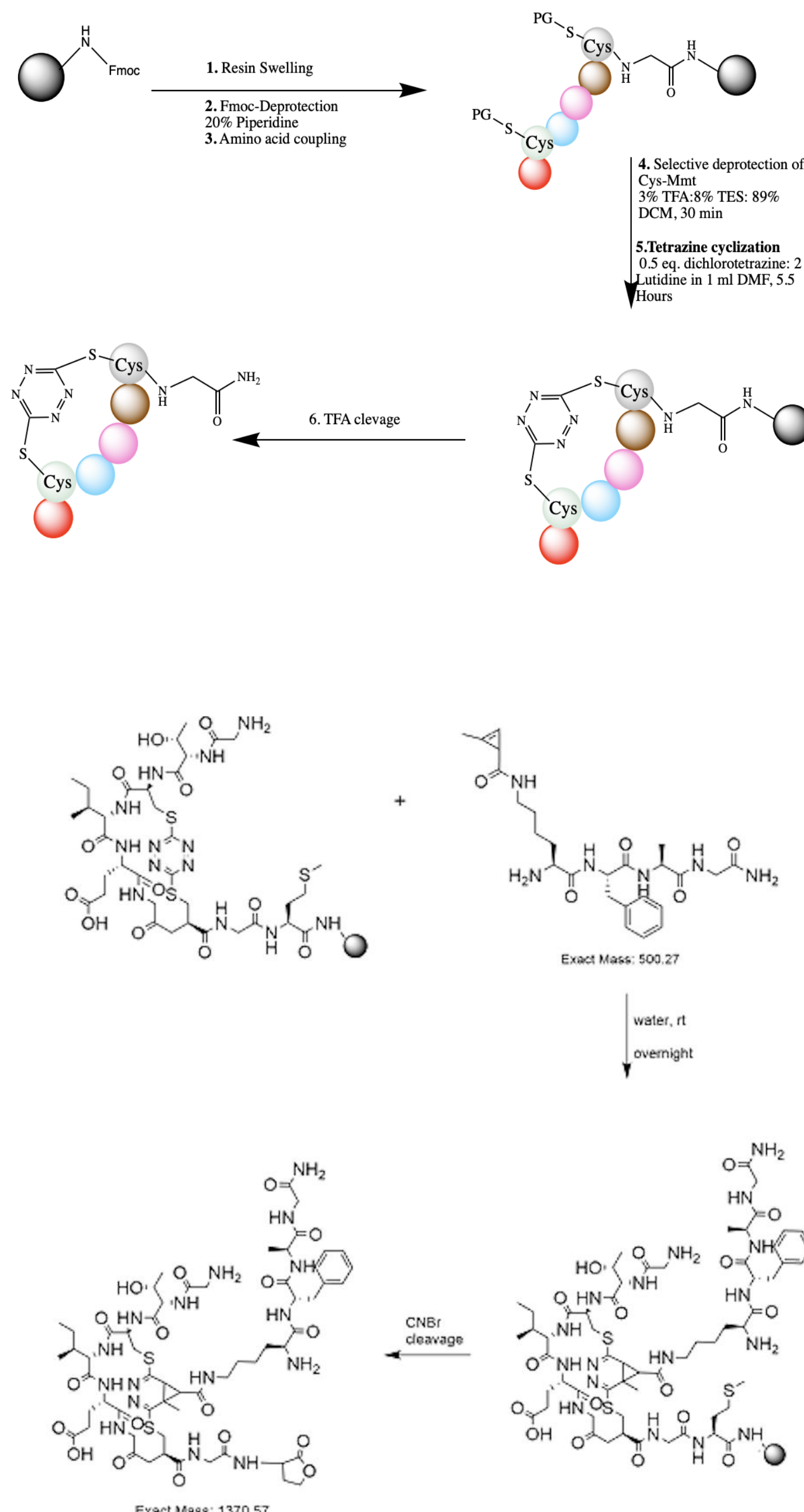
(1) 88% TFA Cleavage (88% TFA, 5% Phenol, 5% Water and 2% TES)

(4) IEDDA: cyclized peptide-tetrazine + Cyclopropene on lysine and a peptide chain

(5) CNBr Cleavage (CNBr + formic acid + water)

(6) HPLC (to purify the compounds)

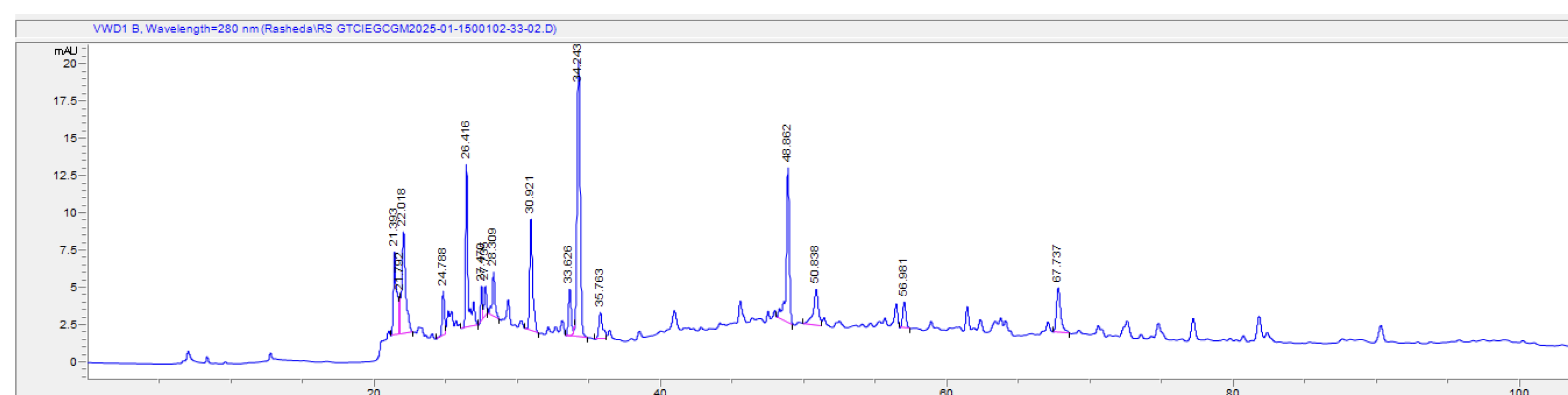
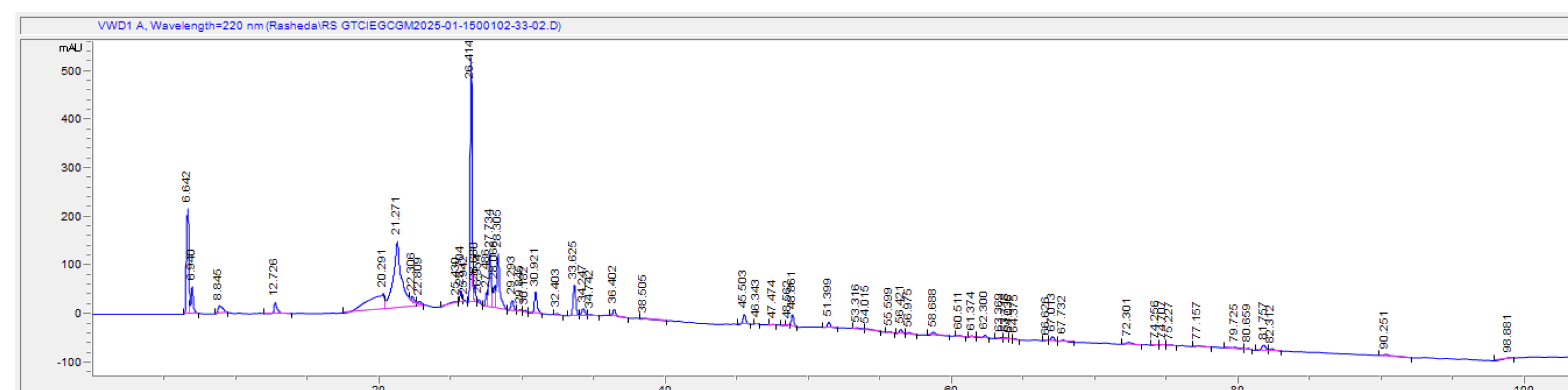
(7) LCMS (to check mass of compounds)



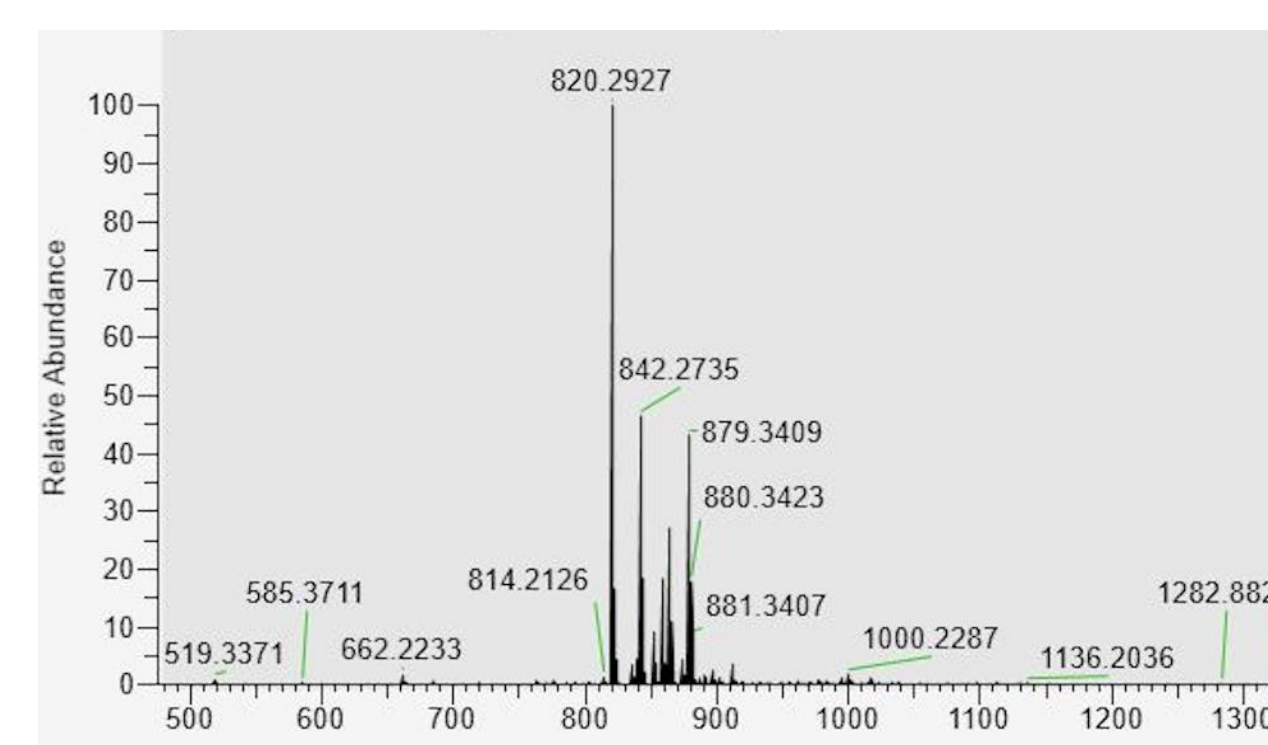
Results

Peptide 1:

HPLC Results



MS results for
linear peptide



MS results for
cyclic peptide

